Bernie Liu

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Education

University of Waterloo Sept. 2026 – Present
Bachelor of Software Engineering (Honours, Co-op) - Presidents Scholarship of Distinction *Waterloo, Ontario*

Experience

Software Engineer – Eve Team 2026 – Present
WATonomous *Waterloo, Ontario*

- Reduced BEVFusion inference latency by **35%** through TensorRT INT8 optimization while maintaining detection accuracy within **2%** of the PyTorch baseline.
- Identified object-tracking, GPU memory, and inference bottlenecks using **50+** ROS bag recordings, improving autonomous vehicle perception pipeline processing speed by **10%**.

Software Engineer Aug. 2026 – Present
UW Cube Design Incubator *Waterloo, Ontario*

- Improve website performance and internal workflows by **15%** by developing full-stack features and internal tools using **Next.js, React, FastAPI, Prisma, and MongoDB**.
- Improve UWCUBE's **AI chatbot response accuracy by 10%** by integrating backend APIs, database services, and application logic to expand knowledge coverage and improve the user experience.

Full-Stack Developer July 2025 – July 2026
Teenage Theory *Richmond Hill, Ontario*

- Co-founded a full-stack blogging platform using Next.js, React, Prisma, and MongoDB, implementing **10+** routes and **5+** database-backed features for publishing, authentication, and content management.
- Built CRUD workflows, responsive interfaces, and Firebase Storage integration for image uploads, supporting **2,000+ site visits** across mobile and desktop devices.
- Optimized MongoDB and Prisma queries through schema refinement, indexing, and reduced redundant database calls, cutting average content retrieval latency by approximately **35%** and reducing database operations by **25%**.

Software Developer Sept. 2023 – June 2026
Richmond Green Computer Science and Model United Nations Club *Richmond Hill, Ontario*

- Developed the official Model UN website using **HTML, CSS, JavaScript, and Tailwind CSS**, serving **300+ users** while leading a team of **10** contributors.
- Reduced invoice preparation time by approximately **80%** by building a **Python and PyQt5** desktop automation system to parse CSV and Excel delegation data and populate invoice records.
- Generated **50+ formatted invoices in under 1 second** using **python-docx**, eliminating repetitive manual data entry and standardizing invoice formatting across Model UN delegations.

Projects

Snitch | *Next.js, TypeScript, FastAPI, Python, Firebase, OpenAI API* Sept. 2026

- Won a **2nd-place prize at Hack the North 2026** by building an AI-assisted interview analysis platform that identified suspicious AI-assisted responses with **80% accuracy** across a labeled test set using latency, linguistic, follow-up, and CV-consistency signals.
- Built a **Next.js** interview interface and **FastAPI** analysis pipeline integrating AssemblyAI, GPTZero, OpenAI, and Firebase, processing recorded interviews in **under 15 seconds** and reducing end-to-end analysis latency by **50%** through parallel API processing.

Airport Search Engine | *TypeScript, HTML/CSS* Sept. 2025 – Nov. 2025

- Built a custom airport search engine capable of searching **100,000+ records in under 0.3 ms** and sorting them in under **5 ms**.
- Developed a **greedy pathfinding algorithm** that minimized distance, travel time, or cost while enforcing aircraft range, landing, and transfer constraints.

Technical Skills

Languages: TypeScript, JavaScript, Python, C++, Java, HTML/CSS, SQL
Frameworks & Libraries: React, Next.js, FastAPI, PyTorch, Three.js, Tailwind CSS, PyQt5, Prisma
Tools & Platforms: Git, TensorRT, MongoDB, Firebase, Vercel, VS Code